Gourab Roy

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About Me

I'm a Machine Learning Researcher and Adjunct Lecturer with a focus on machine learning, deep learning, and computer vision. I've earned gold badges in problem-solving with C++ and Python on HackerRank. I've also published a research paper in biomedical imaging, showing my dedication to making a real impact in my field. As an Adjunct Lecturer, I'm passionate about creating an engaging learning environment by combining my research insights with effective teaching methods. I'm on the lookout for a M.Sc. or PhD position in a challenging academic environment where I can work on solving problems that have a global impact.

RESEARCH INTERESTS

1. Computer Vision, 2. Machine Learning, 3. Deep Learning, 4. Medical Image Analysis, 5. Natural Language Processing

Education

Sylhet Engineering College, Sylhet, Bangladesh — February, 2018 - January, 2023
B.Sc.(Engg.) in Computer Science & Engineering — *CGPA: 3.56/4.00*

Khulna Public College, Khulna, Bangladesh — 2016 - 2017
Higher Secondary Certificate — *GPA: 4.67/5.00*

St. Paul's High School, Mongla, Bagerhat — 2014 - 2015
Secondary School Certificate — *GPA: 5.00/5.00*

Publication

- Ornob, T.R., **Roy, G.** & Hassan, E. (2022). "CovidExpert: A Triplet Siamese Neural Network framework for the detection of COVID-19". Informatics in Medicine Unlocked, 101156, (Jan 13, 2023). DOI: 10.1016/j.imu.2022.101156

Undergrad Thesis

- CerebralNet: A deep learning model to detect major abnormalities from brain CT and MRI scan via segmentation and classification techniques.

Working on & Under Review

- CrackClassification4: Classifying Four Types of Cracks in Structural Surfaces Using CNNs (Under Review)
- Enhancing Concrete Performance: Machine Learning Models for Predicting and Classifying Mechanical Properties (Under Review)
- Optical character recognition for ancient scripts: A case study on Syloti Nagri using deep learning models" *(Under Review)*
- Optimizing Multilingual Summarization: Advanced Corpus Creation and Filtering Approach (Under Review)
- BanglaFont45: A Benchmark Dataset for Transfer Learning in Bangla Font Classification. *(Ongoing)*

Experience

Adjunct Lecturer January 2024 – Present
Sylhet Engineering College *Sylhet, Bangladesh*

Responsible for Taking undergraduate classes, marking assessments, consulting students, Conducting research, and supervising Final year thesis students.
Courses instructed:

- **CSE 705:** Peripheral and Interfacing
- **CSE 706:** Peripheral and Interfacing (Sessional)
- **CSE 505:** Microprocessor & MicroController
- **CSE 506:** Microprocessor & MicroController(Sessional)

Projects

Data Analysis Projects *Python, Numpy, Mathplotlib, Pandas, Jupyter Notebook* Source Code

* Developed expertise in exploratory data analysis and visualization by completing diverse projects, including analysis of **Boston housing data, Covid-19 global health outcomes, and Google Play app ratings data**
* Created a Git repository featuring **8 exploratory data analysis projects**, demonstrating skills in data visualization and analysis

Python Projects *Python, PyCharm, Jupyter Notebook* Source Code

* Made some Python projects featuring multiple games and programs, including **Calculator, Dice rolling simulator, Tic Tac Toe game, Caterpillar game, Egg Catcher game, and Hangmarker Game**

Java Application *Java, Netbeans IDE, Mysql, JFrame* Source Code

* Designed and developed a **Billing System application** using **Java and MySQL** for my university project, showcasing skills in programming, database design, and software development

Web Based Project *Html, CSS, JavaScript* Source Code

* Designed and developed a simple Website called Pioneer Bank using **HTML, CSS, JavaScript**
* Added the project in **Github**

Technical Skills

Languages	:	C, C++, Python, Java, JavaScript
ML Libraries	:	PyTorch, Tensorflow, Scikit-learn
Databases	:	MySQL, SQLite
Dev Tools	:	Visual Studio Code, PyCharm, Netbeans IDE
Others	:	Git, Windows, Linux, LaTeX

STANDARDIZED TEST SCORE

IELTS, overall score **6.5** (Listening: 7, Reading: 6.5, Speaking: 6, Writing: 6)

Certifications & Achievements

- Solved 350+ problems in various online judges (e.g., **HackerRank**, **Codeforces**, **UVA**)
- Earned Gold Badges in C++ and Python problem solving on HackerRank
- Completed Machine Learning Specialization course from Coursera
- Completed Data Structures course from Coursera
- Completed Data Visualization in Python Masterclass course from Udemy
- Review contribution to the Informatics in Medicine Unlocked

References

Enamul Hassan
Designation: Assistant Professor,
Department of Computer Science
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Shahjalal University of Science
and Technology.
Teaching Assistant:
Stony Brook University,
Stony Brook, New York, US.
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